

S Block**JEE Main 2020 Chapterwise****Questions with Answer Keys****Chemistry****Q1 JEE Main 2020 - 2 September (Morning)**

The metal mainly used in devising photoelectric cells is

- (A) Na
- (B) Rb
- (C) Li
- (D) Cs

Q2 JEE Main 2020 - 2 September (Evening)

Two elements A and B have similar chemical properties. They don't form solid hydrogencarbonates, but react with nitrogen to form nitrides. A and B, respectively, are

- (A) Li and Mg
- (B) Cs and Ba
- (C) Na and Rb
- (D) Na and Ca

Q3 JEE Main 2020 - 4 September (Morning)

On combustion of Li, Na and K in excess of air, the major oxides formed, respectively, are

- (A) Li_2O , Na_2O_2 and K_2O
- (B) Li_2O_2 , Na_2O_2 and K_2O_2
- (C) Li_2O , Na_2O_2 and KO_2
- (D) Li_2O , Na_2O and K_2O_2

Q4 JEE Main 2020 - 4 September (Evening)

An alkaline earth metal 'M' readily forms water soluble sulphate and water insoluble hydroxide. Its oxide MO is very stable to heat and does not have rock-salt structure. M is

- (A) Ca
- (B) Mg
- (C) Sr
- (D) Be

Q5 JEE Main 2020 - 6 September (Morning)

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Among the sulphates of alkaline earth metals, the solubilities of BeSO_4 and MgSO_4 in water, respectively, are

- (A) poor and poor
- (B) poor and high
- (C) high and poor
- (D) high and high

Q6 JEE Main 2020 - 6 September (Evening)

Match the following compounds (Column-I) with their uses (Column-II)

S.No.	Column-I	S.No.	Column-II
(I)	Ca(OH)_2	(A)	casts of statues
(II)	NaCl	(B)	white wash
(III)	$\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}$	(C)	antacid
(IV)	CaCO_3	(D)	washing soda preparation

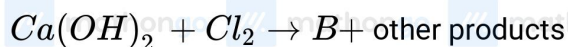
- (A) (I) – (B), (II) – (C), (III) – (D), (IV) – (A)
- (B) (I) – (B), (II) – (D), (III) – (A), (IV) – (C)
- (C) (I) – (C), (II) – (D), (III) – (B), (IV) – (A)
- (D) (I) – (D), (II) – (A), (III) – (C), (IV) – (B)

Q7 JEE Main 2020 - 7 January (Evening)

$\text{NaOH} + \text{Cl}_2 \rightarrow \text{A} + \text{other products}$
Hot & conc.

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S Block**JEE Main 2020 Chapterwise****Questions with Answer Keys****Chemistry**

Cold & dil.

A & *B* are respectively

- (A) $NaClO_3$, $Ca(OCl)_2$
- (B) $NaClO_3$, $Ca(ClO_3)_2$
- (C) $NaCl$, $Ca(ClO_3)_2$
- (D) $NaClO$, $Ca(ClO_3)_2$

Q8 JEE Main 2020 - 8 January (Morning)

When gypsum is heated to 393 K it forms

- (A) dead burnt plaster
- (B) Anhydrous $CaSO_4$
- (C) $CaSO_4 \cdot \frac{1}{2}H_2O$
- (D) $CaSO_4 \cdot 5H_2O$

Q9 JEE Main 2020 - 8 January (Evening)

A metal (*A*) on heating in nitrogen gas gives compound *B*. *B* on treatment with H_2O gives a colorless gas which when passed through $CuSO_4$ solution gives a dark blue-violet coloured solution. *A* and *B* respectively, are :

- (A) Mg , Mg_3N_2
- (B) Na , Na_3N
- (C) Mg , $Mg(NO_3)_2$
- (D) Na , $NaNO_3$

Q10 JEE Main 2020 - 9 January (Evening)

The reaction of $H_3N_3B_3Cl_3$ (*A*) with $LiBH_4$ in tetrahydrofuran gives inorganic benzene (*B*).

Further, the reaction of (*A*) with (*C*) leads to $H_3N_3B_3(Me)_3$. Compounds (*B*) and (*C*) respectively, are:

- (A) Boron nitride and $MeBr$
- (B) Borazine and $MeMgBr$
- (C) Borazine and $MeBr$
- (D) Diborane and $MeMgBr$

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Questions with Answer Keys

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Chemistry

Q11 JEE Main 2020 - 9 January (Evening)

Among the statements (a)-(d), the correct ones are :

- (a) Lithium has the highest hydration enthalpy among the alkali metals.
- (b) Lithium chloride is insoluble in pyridine.
- (c) Lithium cannot form ethynide upon its reaction with ethyne.
- (d) Both lithium and magnesium react slowly with H_2O .

- (A) (a), (b) and (d) only
- (B) (b) and (c) only
- (C) (a) and (d) only
- (D) (a), (c) and (d) only

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Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

Answer Key

Q1 (D)

Q2 (A)

Q3 (C)

Q4 (D)

Q5 (D)

Q6 (B)

Q7 (A)

Q8 (C)

Q9 (A)

Q10 (B)

Q11 (D)

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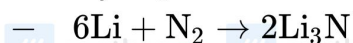
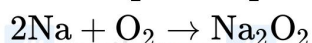
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S Block**JEE Main 2020 Chapterwise****Hints & Solutions****Chemistry****Q1**

Cs has least work function value because of less electronegativity that's why it is used in photoelectric cells.

Q2

Solid hydrogencarbonates are not formed by lithium and magnesium.

**Q3****Q4**

Solubility of BeSO_4 is highest among the given $\text{BeSO}_4 > \text{MgSO}_4 > \text{CaSO}_4 > \text{SrSO}_4 > \text{BaSO}_4$

BeSO_4 is soluble and $\text{Be}(\text{OH})_2$ is insoluble. Solubility order for hydroxide.



BeO does not form rock salt like structure.

Q5

BeSO_4 and MgSO_4 are readily soluble in water.

Reason $\rightarrow$ The greater hydration enthalpies of Be^{2+} and Mg^{2+} ions dominate over their lattice enthalpies and therefore their sulphates are highly soluble.

Q6

$\text{Ca}(\text{OH})_2$ — Used in white wash NaCl — Used for the preparation of washing soda (Na_2CO_3)

$\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ — Used for making casts of statues

CaCO_3 — Used as an antacid

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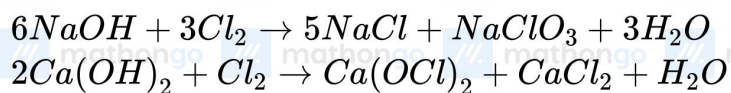
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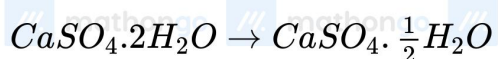
Hints & Solutions

Chemistry

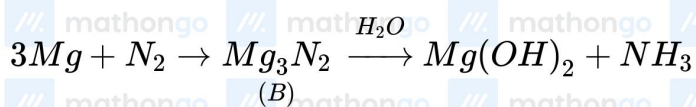
Q7



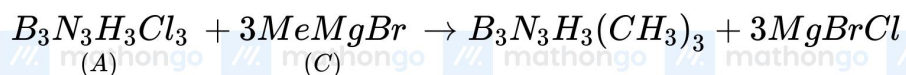
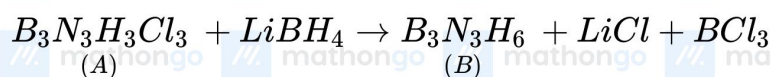
Q8



Q9



Q10



Q11

Theory based